# Sampling: The Bridge from Continuous to Discrete

Chapter 3

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# Outline

- Sampling Theorem
- Signal Reconstruction
- Practical Difficulty in Sampling and Reconstruction
- Sampling of Bandpass Signals
- Spectral Sampling Theorem
- Analog to Digital Conversion
- Digital to Analog Conversion

# 3.1 & 3.2
# Sampling and Signal Reconstruction

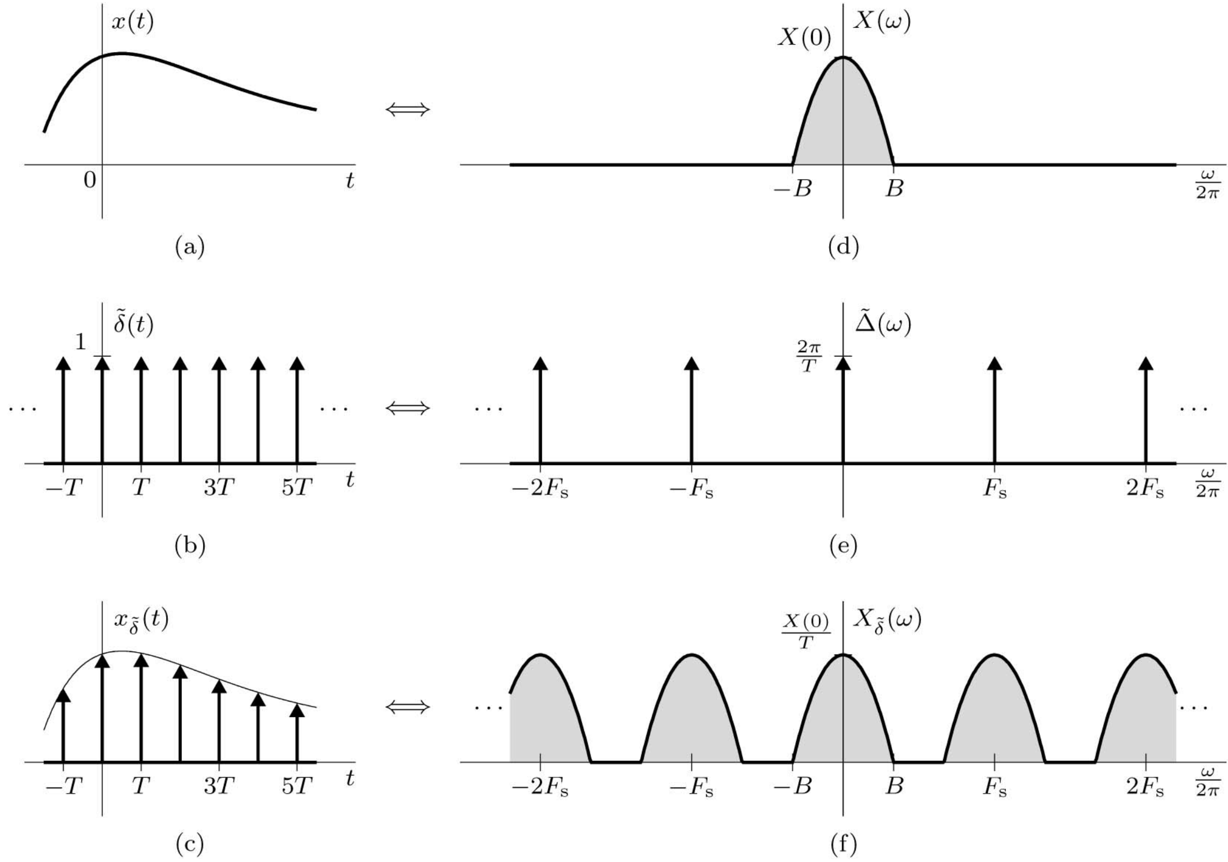
# Sampling Theorem

A real signal whose spectrum is bandlimited to $B$ Hz [$X(f) = 0$ for $|f| > B$ ] can be reconstructed exactly from its samples taken uniformly at a rate $F_s > 2B$ samples per second. When $F_s = 2B$ then $F_s$ is the Nyquist rate.

$$\tilde{\delta}(t) = \sum_{k=-\infty}^{k=\infty} \delta(t - kT_s) = \sum_{k=-\infty}^{\infty} \frac{1}{T_s} e^{jk\omega_s t}$$

$$x_{\tilde{\delta}}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T_s} e^{jk\omega_s t}\, x(t)$$

$$X_{\tilde{\delta}}(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{k=\infty} X(\omega - k\omega_s)$$



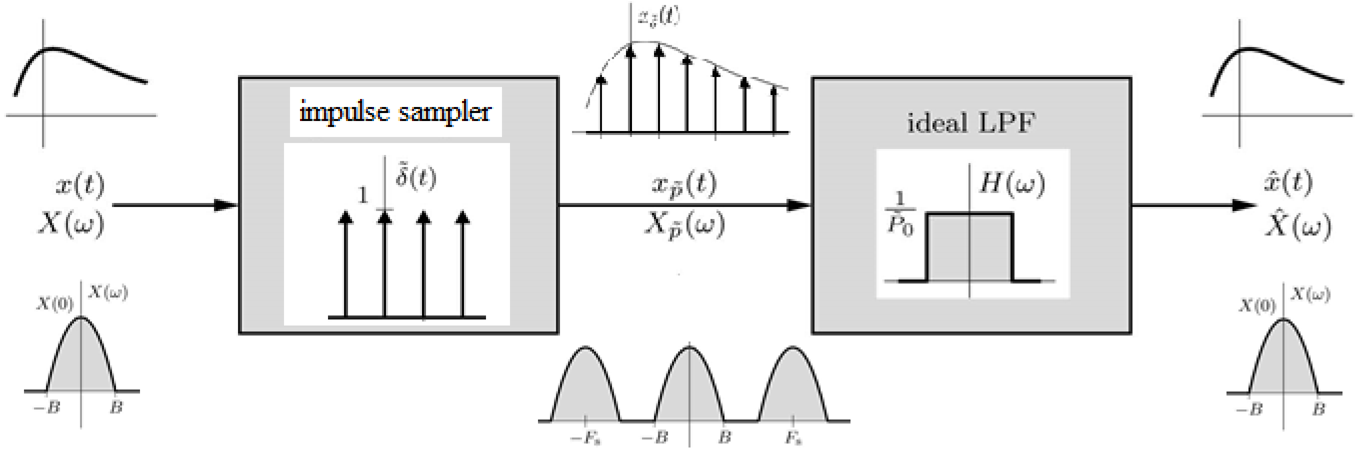
# Ideal Signal Reconstruction from Samples

## In Frequency Domain

$$X_{\tilde{\delta}}(\omega) = \frac{1}{T_s}\sum_{k=-\infty}^{\infty} X(\omega - k\omega_s)$$

$$H(\omega) = T_s \Pi\left(\frac{\omega}{\omega_s}\right)$$

$$\hat{X}(\omega) = X_{\tilde{\delta}}(\omega)H(\omega)$$



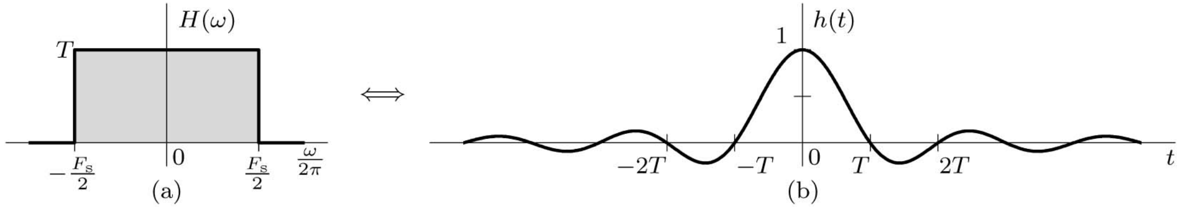

## In Time Domain

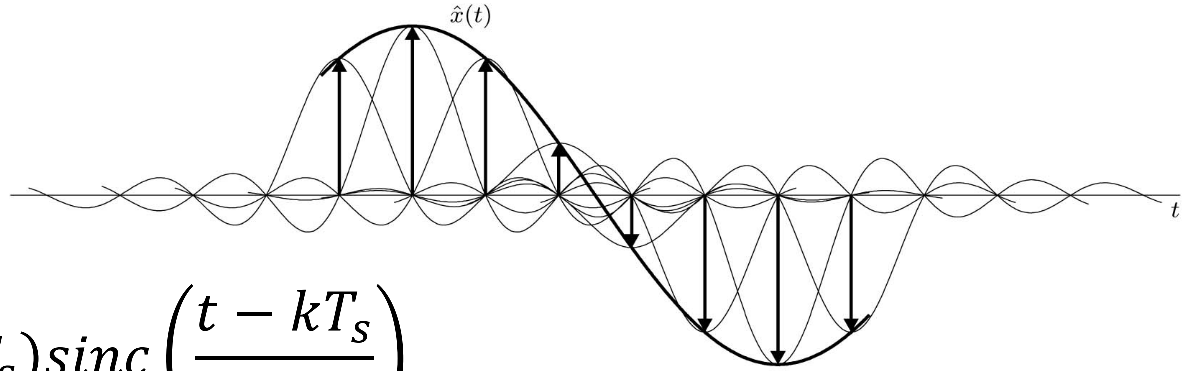
$$x_{\tilde{\delta}}(t) = \sum_{k=-\infty}^{\infty} x(kT_s)\delta(t - kT_s)$$

$$h(t) = \frac{1}{2\pi} sinc(t/2\pi T_s)$$

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} x(kT_s)h(t - kT_s)$$

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{2\pi} x(kT_s) sinc\left(\frac{t - kT_s}{2\pi T_s}\right)$$



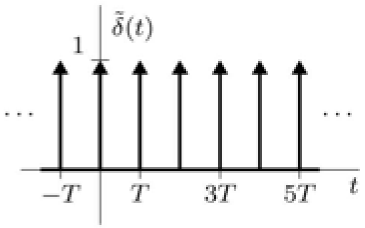

$$\frac{B}{\pi}\text{sinc}\left(\frac{Bt}{\pi}\right) \longleftrightarrow \Pi\left(\frac{\omega}{2B}\right)$$

# Review of Pulse Train

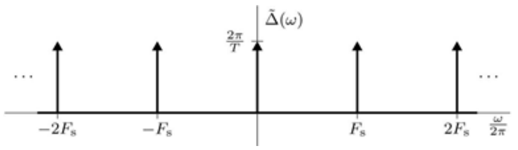
$$e^{jk\omega_s t} \overset{\text{FT}}{\longleftrightarrow} 2\pi\delta(\omega - k\omega_s)$$

## Ideal sampling impulses

$$\tilde{\delta}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T_s} e^{jk\omega_s t}$$

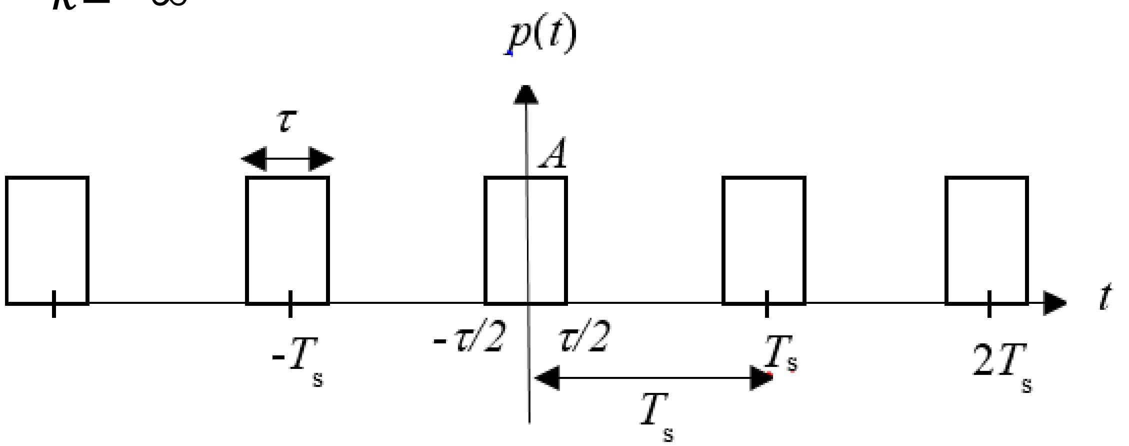



$$\tilde{\delta}(\omega) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{T_s} \delta(\omega - k\omega_s)$$

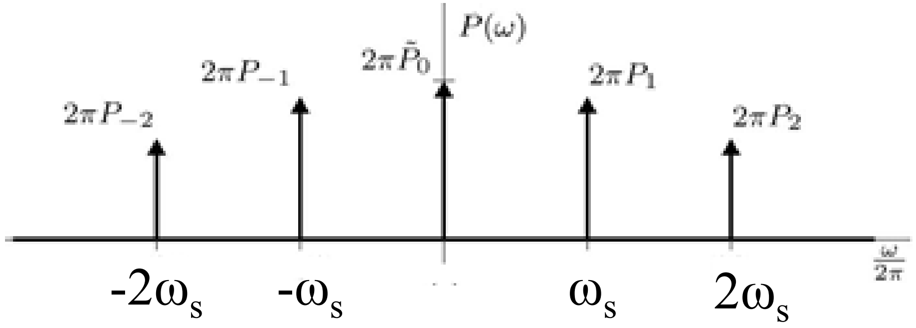



## Practical sampling pulses

$$p(t) = \sum_{k=-\infty}^{\infty} \underbrace{\frac{A\tau}{T_s} \operatorname{sinc}\left(\frac{k\omega_s \tau}{2\pi}\right)}_{X_k} e^{jk\omega_s t}$$




$$P(\omega) = \sum_{k=-\infty}^{\infty} \underbrace{\frac{2\pi A\tau}{T_s} \operatorname{sinc}\left(\frac{k\omega_s \tau}{2\pi}\right)}_{P_k} \delta(\omega - k\omega_s)$$



# Practical Sampling

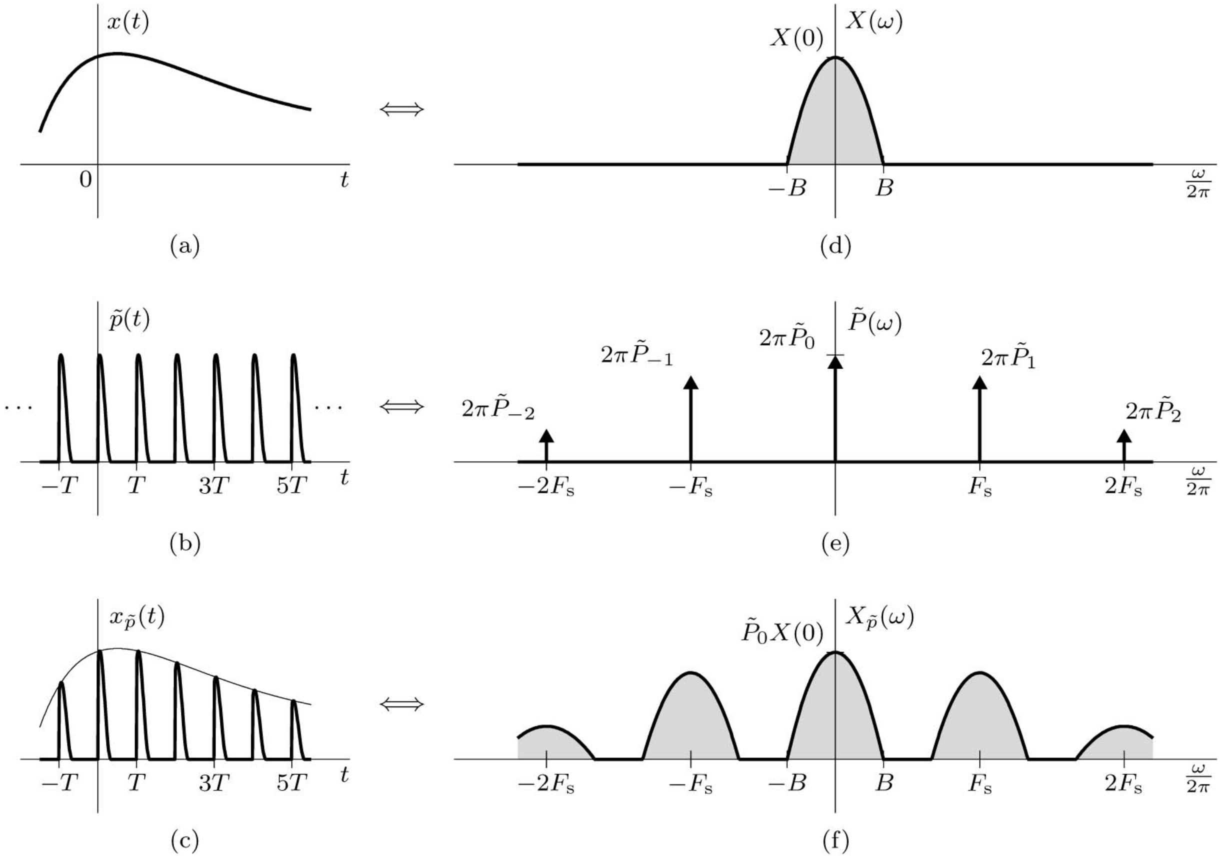
In practice we sample the signal $x(t)$ by multiplying it by a $T$-periodic train $\tilde{p}(t)$.

$$\tilde{p}(t) = \sum_{k=-\infty}^{k=\infty} p(t - kT_s)$$

$$\tilde{p}(t) = \sum_{k=-\infty}^{\infty} \tilde{P}_k e^{jk\omega_s t}$$

$$x_{\tilde{p}}(t) = \tilde{p}(t)x(t) = \sum_{k=-\infty}^{\infty} \tilde{P}_k e^{jk\omega_s t} x(t)$$

$$X_{\tilde{p}}(\omega) = \sum_{k=-\infty}^{k=\infty} \tilde{P}_k X(\omega - k\omega_s)$$



# Example of Practical Sampling

Consider the signal $x(t) = \text{sinc}^2(5t)$ and the pulse train $\tilde{p}(t) = \sum_{n=-\infty}^{\infty} \Pi(\frac{t-0.1n}{0.025})$. Plot the pulse-sampled signal $x_{\tilde{p}}(t) = \tilde{p}(t)x(t)$ and its spectrum $X_{\tilde{p}}(\omega)$. Is $x_{\tilde{p}}(t)$ sampled above, below, or at the Nyquist rate? If $x_{\tilde{p}}(t)$ is applied to a unit-gain ideal lowpass filter with a bandwidth of 5 Hz, what is the output signal $y(t)$?

$\frac{B}{2\pi}\text{sinc}^2\left(\frac{Bt}{2\pi}\right)$ $\quad$ $\Lambda\left(\frac{\omega}{2B}\right)$

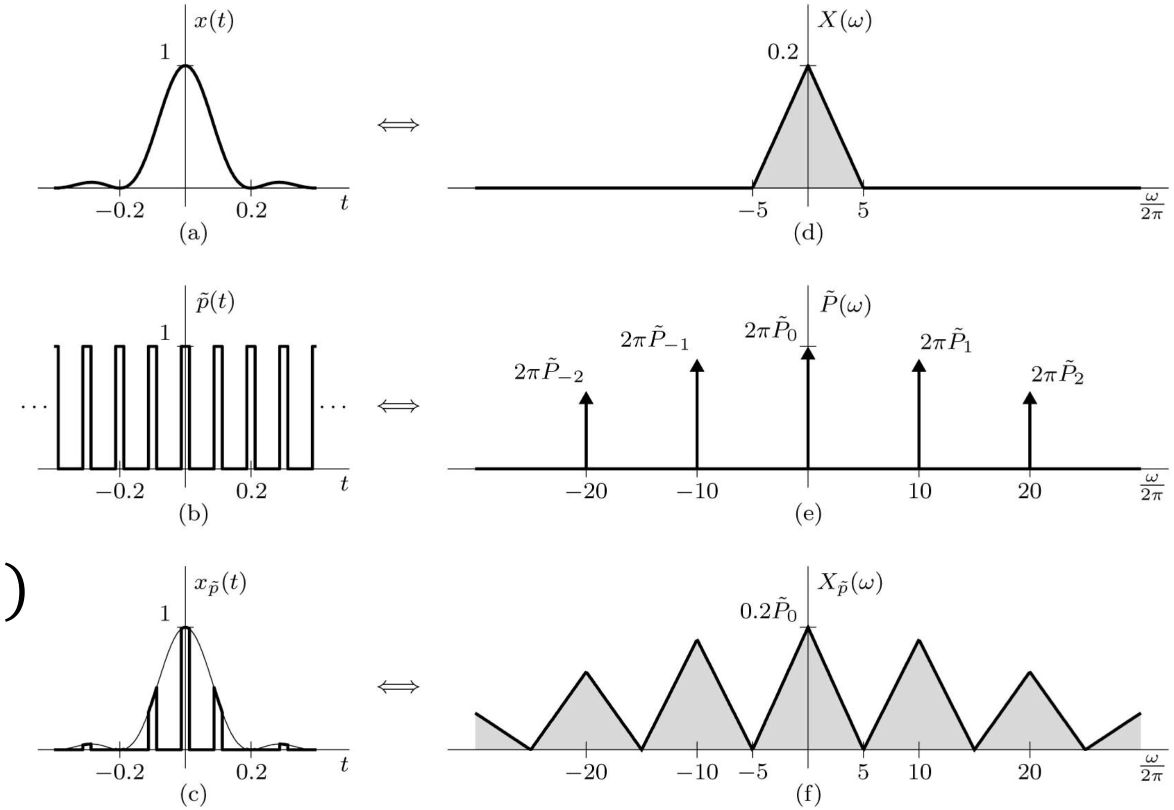
$$X(\omega) = 0.2\Lambda(\tfrac{\omega}{20\pi})$$

$$\tilde{p}(t) = \sum_{k=-\infty}^{\infty} \tilde{P}_k e^{jk\omega_s t} = \sum_{k=-\infty}^{\infty} \frac{1}{4}\text{sinc}(k/4)e^{jk\omega_s t}$$

$$x_{\tilde{p}}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{4} sinc(k/4)e^{jk\omega_s t}\, x(t)$$

$$X_{\tilde{p}}(\omega) = \sum_{k=-\infty}^{k=\infty} \frac{1}{4} sinc(k/4)X(\omega - k\omega_s)$$

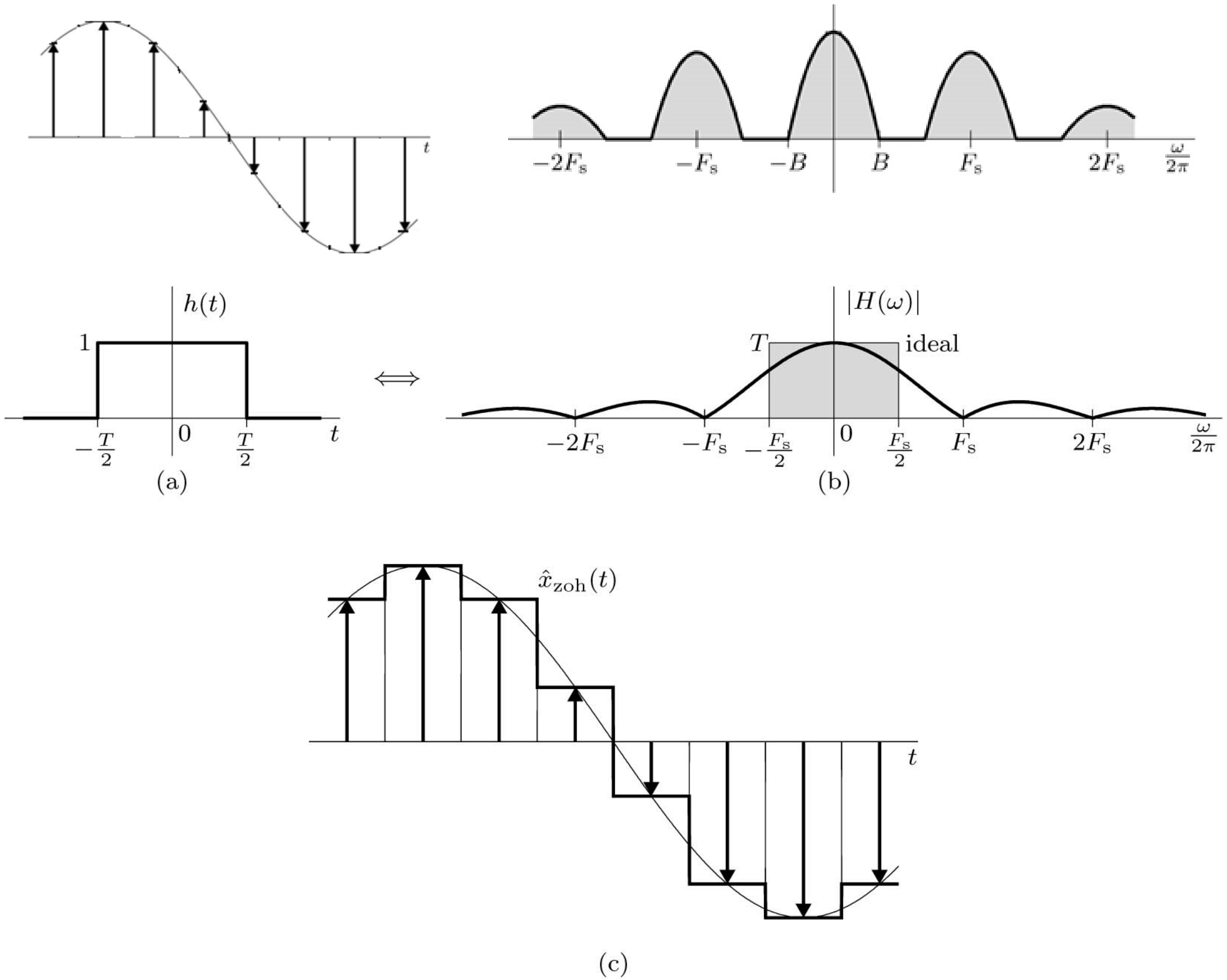
$$y(t) = \frac{1}{4}x(t)$$



# Practical Interpolation

## The Zero-Order Hold Filter

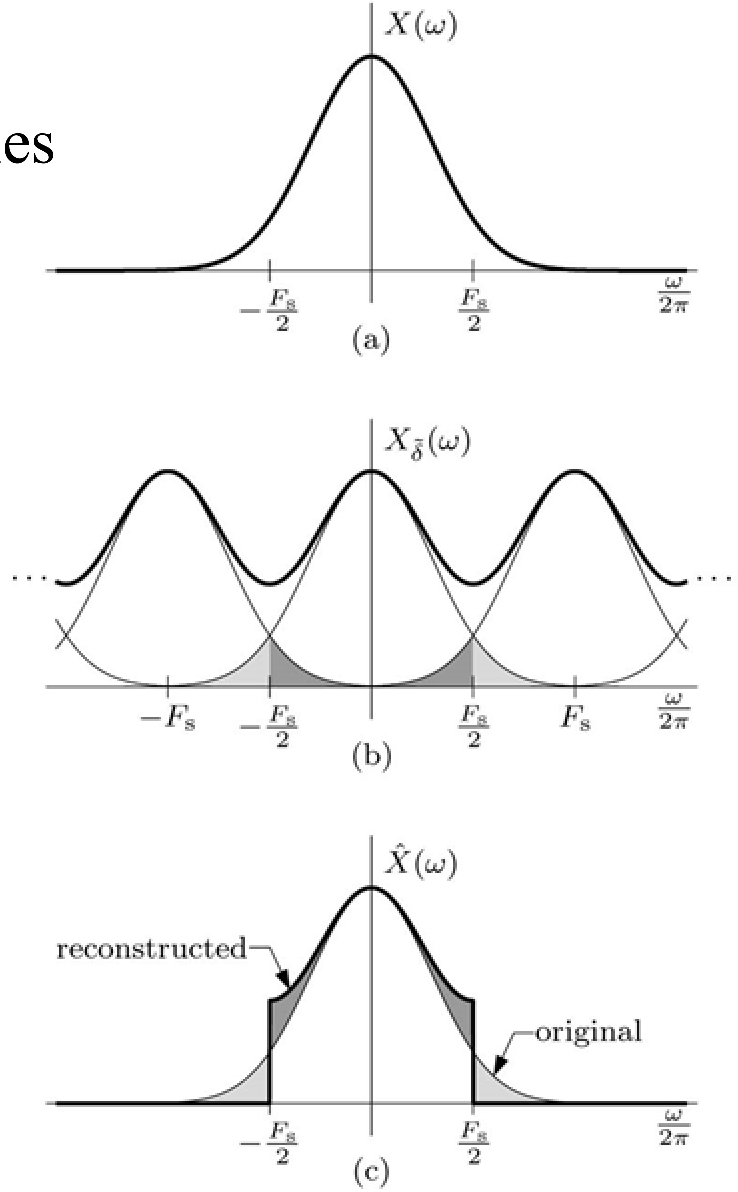
$$H(\omega) = T\,\mathrm{sinc}\left(\frac{\omega T}{2\pi}\right)$$



# 3.3
# Practical Difficulties in Sampling

# Practical Difficulty in Sampling and Reconstruction

Aliasing: If a continuous time signal is sampled below the Nyquist rate then some of the high frequencies will appear as low frequencies and the original signal can not be recovered from the samples.

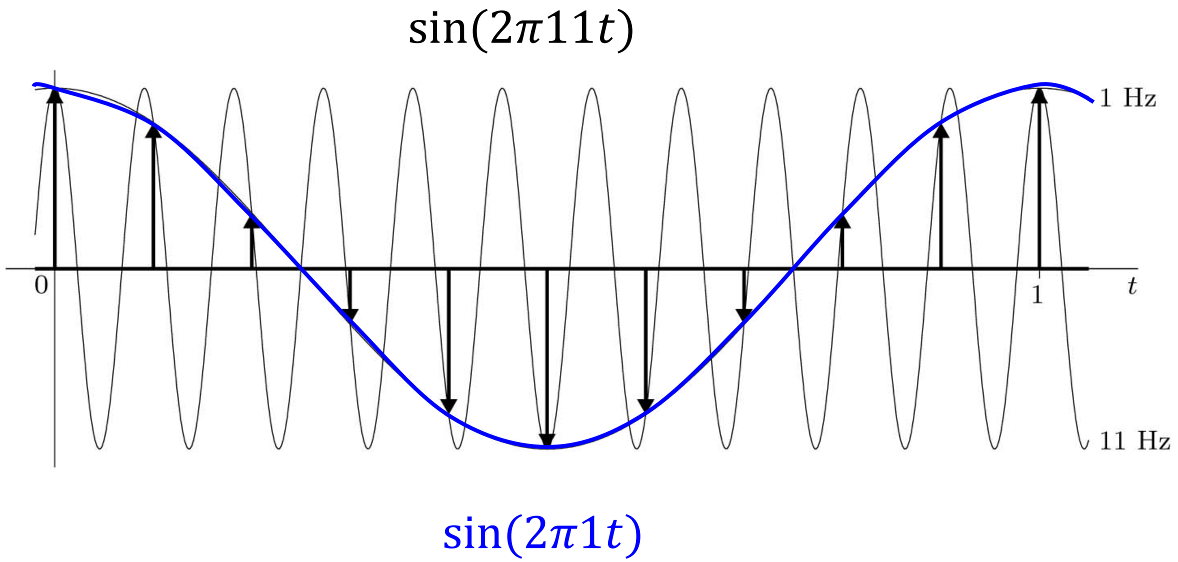


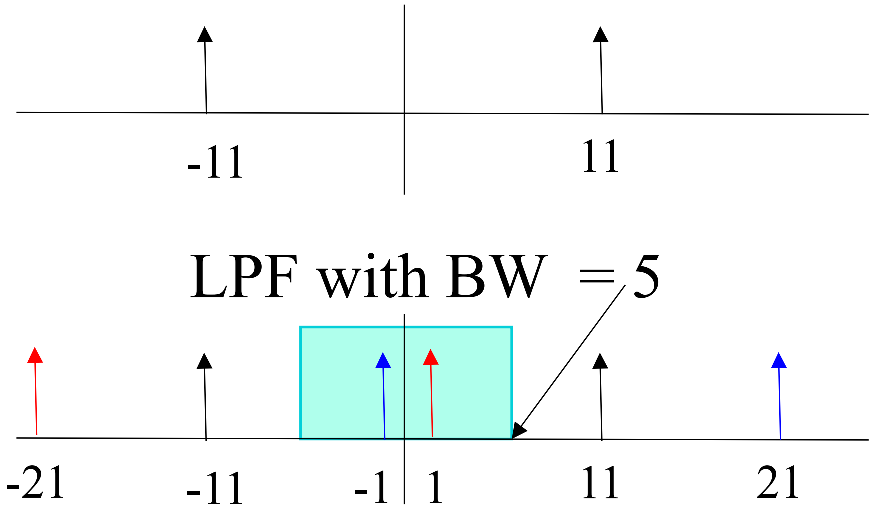

For the spectra of real signals, the tail appears to fold or invert about $F_s/2$. Thus, a component of frequency $F_s/2 + f_1$ shows up as or impersonates a component of lower frequency $F_s/2 - f_1$ in the reconstructed signal. This causes distortion of low-frequency ($|f| < F_s/2$) signal content. The frequency $F_s/2$ is called the *folding frequency*, and the tail inversion is known as *spectral folding* or *aliasing.*

If a signal is time limited, it cannot be bandlimited, and vice versa.

# Aliasing in Sinusoids

A sinusoid with 11 Hz frequency sampled with sampling rate $F_s = 10$ z.



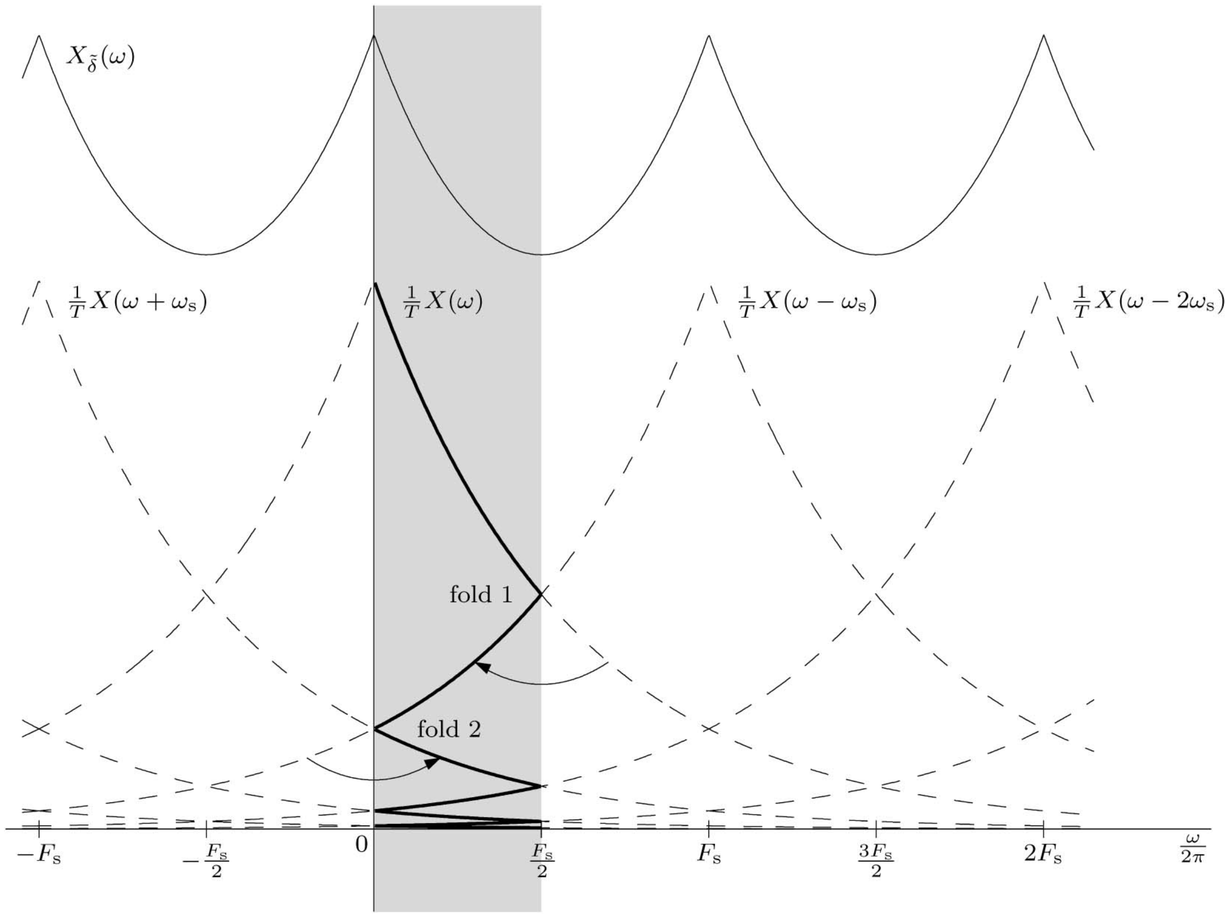



# Multiple Aliasing

The *apparent frequency* $f_a$ that is in the fundamental band ($-F_s/2 \leq f_a \leq F_s/2$) of undersampled signal of frequency $f_0$ is.

$$f_a = \left| \langle f_0 + F_s/2 \rangle_{F_s} - F_s/2 \right|$$

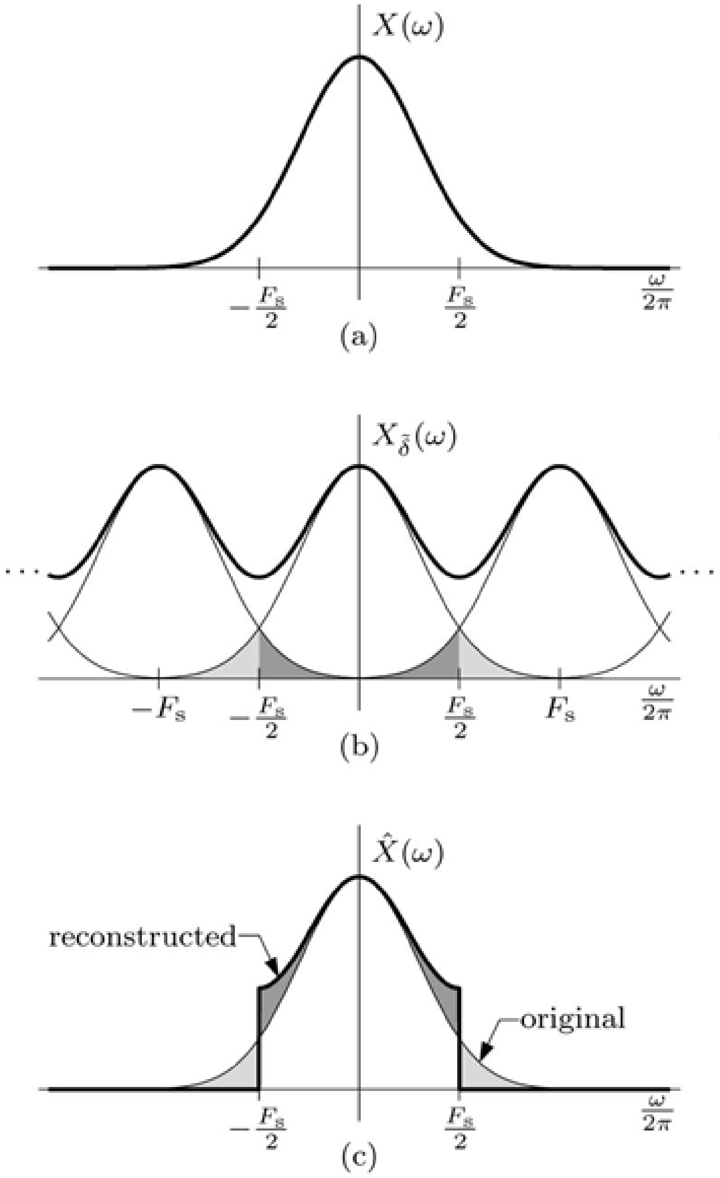
where $\langle a \rangle_b$ is the *modulo operation a* modulo *b*.



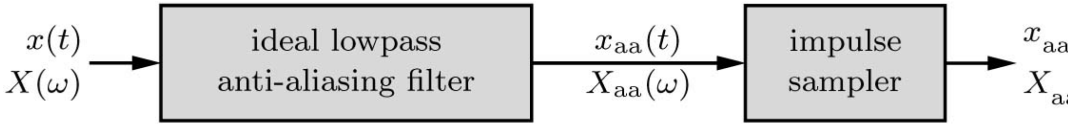

Example:

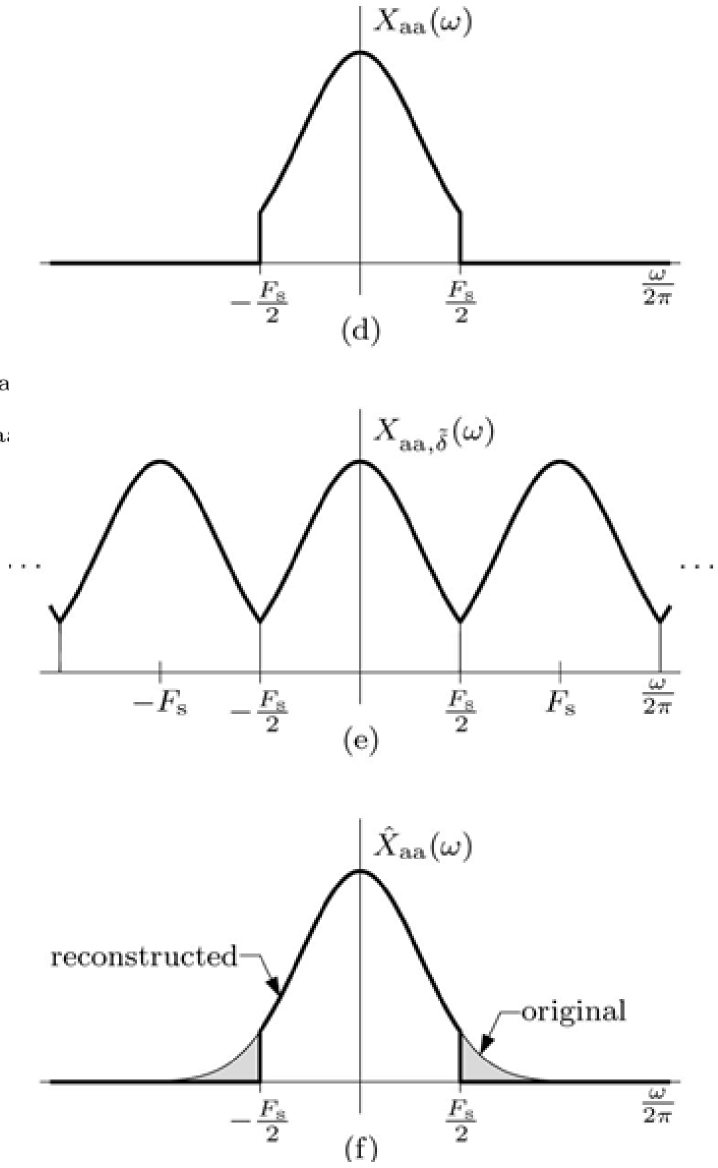
What are the apparent frequencies of sinusoidals with 3, 8, 23 Hz frequency sampled with sampling rate $F_s$ = 10 z.

# Reduce the Effect of Aliasing




## Anti-Aliasing Filter

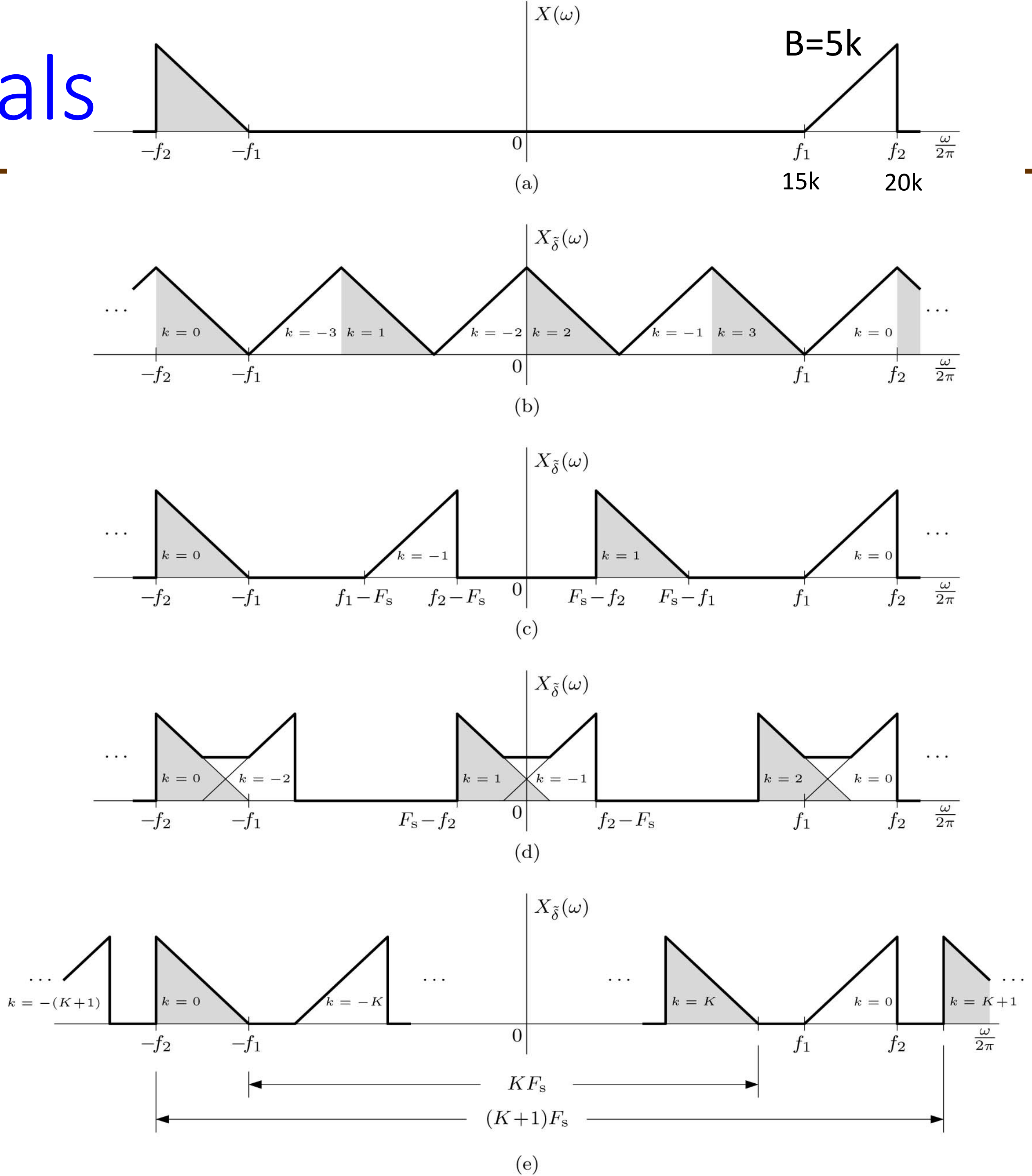



HW 1 is due

# 3.4
# Sampling of Bandpass Signals

# Sampling of Bandpass Signals




Sampling a bandwidth $B$ bandpass signal:

(a) original spectrum,

(b) $f_2 = 4B$ and $F_s = 2B$,

(c) $f_2 = 4B$ and $F_s = 4.75B$,

(d) $f_2 = 4B$ and $F_s = 3.25B$,

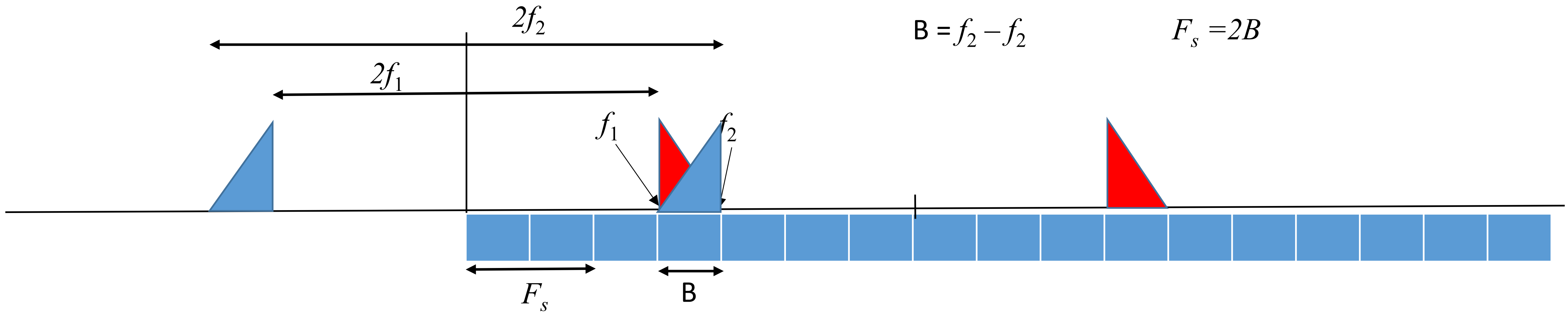
(e) general sampling case

$$KF_s < 2f_1 \quad \text{and} \quad (K+1)F_s > 2f_2$$

$$\frac{2f_2}{K+1} < F_s < \frac{2f_1}{K} \qquad K = 0, 1, 2, \ \ldots$$

$$\frac{2f_2}{K+1} < F_s < \frac{2(f_2 - B)}{K}$$

$$\frac{2(f_2 - B)}{K} > \frac{2f_2}{K+1} \qquad K_{\max} = \left\lfloor \frac{f_2}{B} \right\rfloor - 1$$




Sampling a bandwidth $B$ bandpass signal:
(a) original spectrum,
(b) $f_2 = 4B$ and $F_s = 2B$,
(c) $f_2 = 4B$ and $F_s = 4.5B$,
(d) $f_2 = 4B$ and $F_s = 3.5B$,

$$KF_s < 2f_1 \quad \text{and} \quad (K+1)F_s > 2f_2$$

$$\frac{2f_2}{K+1} < F_s < \frac{2f_1}{K}$$

$$\frac{2(f_2 - B)}{K} > \frac{2f_2}{K+1}$$

$$K_{\max} = \left\lfloor \frac{f_2}{B} \right\rfloor - 1$$

Example 3.6: Determine the permissible range of the uniform sampling rate for a bandpass signal with $f_1 = 15$ kHz and $f_2 = 20$ kHz.

# 3.5
# Spectral Sampling Theorem

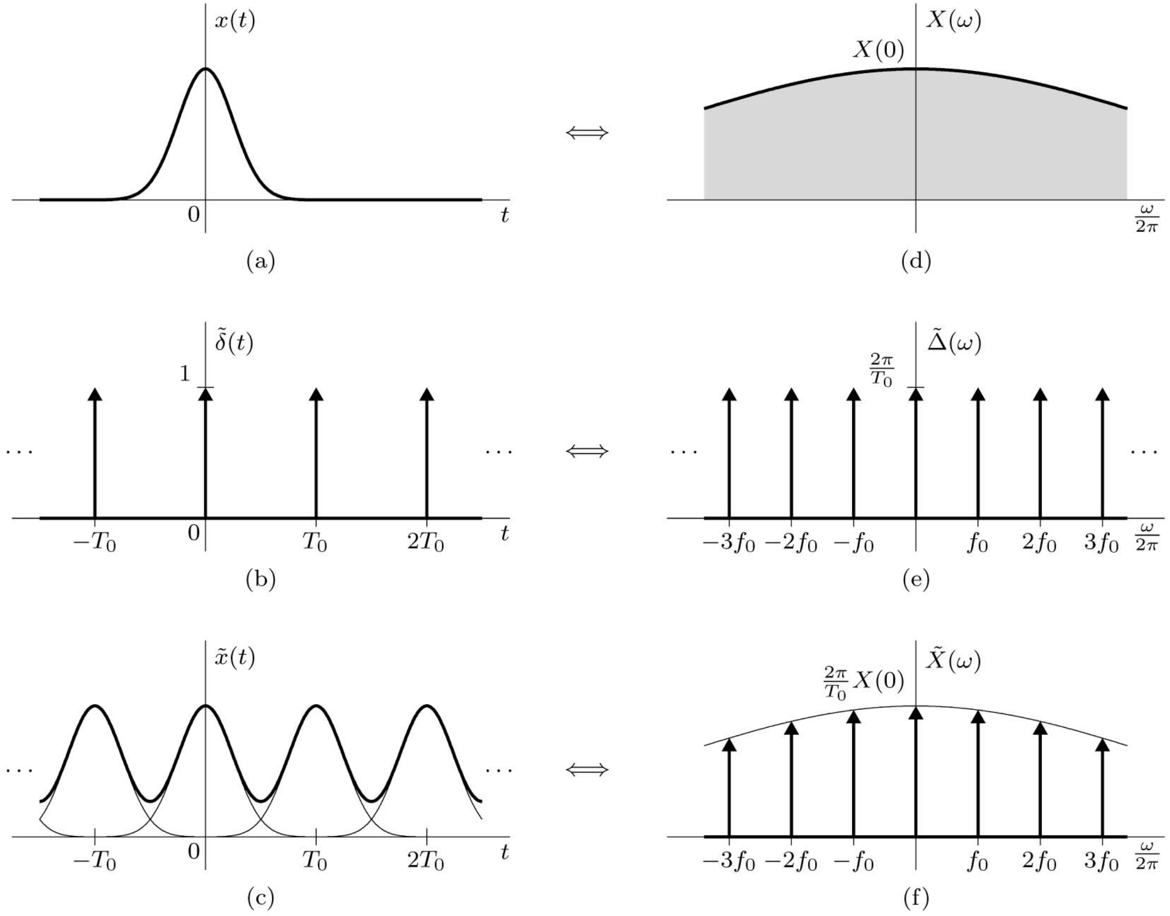
Skip section 3.5: will be covered in DFT

# The Spectral Sampling Theorem

The *spectral sampling theorem* states that the spectrum $X(\omega)$ of a signal $x(t)$ timelimited to a duration of $T_o$ seconds can be reconstructed from the samples of $X(\omega)$ taken at a rate $R$ samples/Hz, where $R \geq T_o$ (the signal width or duration) is in seconds. The frequency resolution $f_o = 1/\, T_o$.

$$\tilde{X}(\omega) = X(\omega)\tilde{\Delta}(\omega) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{T_0} X(k\omega_0)\delta(\omega - k\omega_0)$$

$$\tilde{x}(t) = x(t) * \tilde{\delta}(t) = \sum_{n=-\infty}^{\infty} x(t - nT_0)$$



# Reconstructing $x(t)$ from Spectral Samples

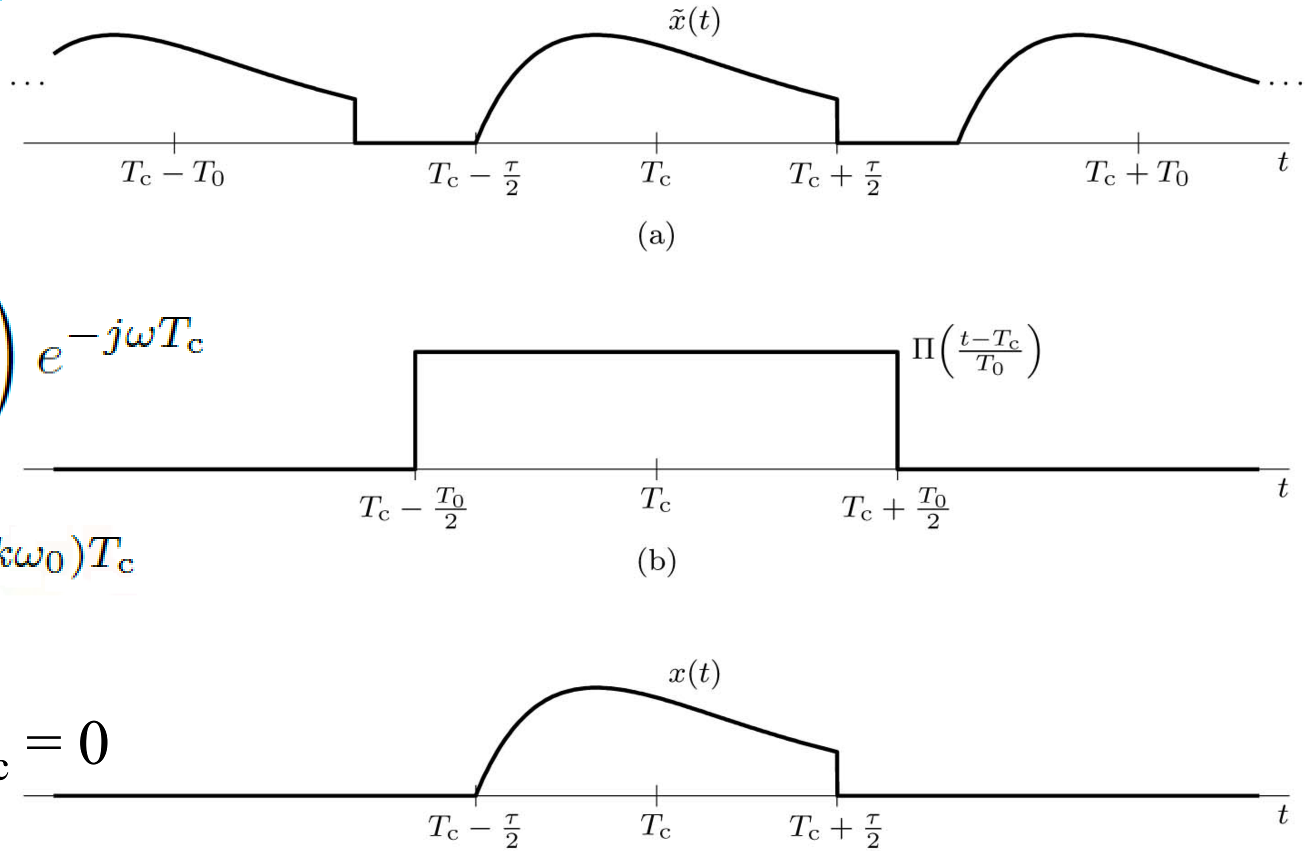
$$x(t) = \tilde{x}(t)\Pi\left(\frac{t-T_c}{T_0}\right)$$

$$\Pi\left(\frac{t-T_c}{T_0}\right) \Longleftrightarrow T_0 \mathrm{sinc}\left(\frac{\omega T_0}{2\pi}\right) e^{-j\omega T_c}$$

$$X(\omega) = \frac{1}{2\pi}\tilde{X}(\omega) * T_0 \mathrm{sinc}\left(\frac{\omega T_0}{2\pi}\right) e^{-j\omega T_c}$$

$$X(\omega) = \sum_{k=-\infty}^{\infty} X(k\omega_0)\delta(\omega - k\omega_0) * \mathrm{sinc}\left(\frac{\omega T_0}{2\pi}\right) e^{-j\omega T_c}$$

$$X(\omega) = \sum_{k=-\infty}^{\infty} X(k\omega_0)\mathrm{sinc}\left(\frac{\omega T_0}{2\pi} - k\right) e^{-j(\omega - k\omega_0)T_c}$$




If the pulse $x(t)$ is centered at the origin, then $T_c = 0$

Example 3.7: Find the unit duration signal $x(t)$, centered at the origin, whose spectrum $X(\omega)$, when sampled at intervals of $f_o = 1$ Hz (the Nyquist rate), is $X(k2\pi f_o) = 1$ for $k = 0$ and zero otherwise.

# 3.6
# Analog to Digital Conversion

# 3.6 Analog to Digital Conversion (ADC)

ADC

01110011 11001101 01011100

# 3.6 Analog to Digital Conversion

## Sampling and Linear Quantization

Analog signal: Its amplitude can take on any value over a continuous range.
Digital signal: Its amplitude can take on only a finite number of values.

# 3.6 Analog to Digital Conversion

## Linear Quantization

How many bits per sample?

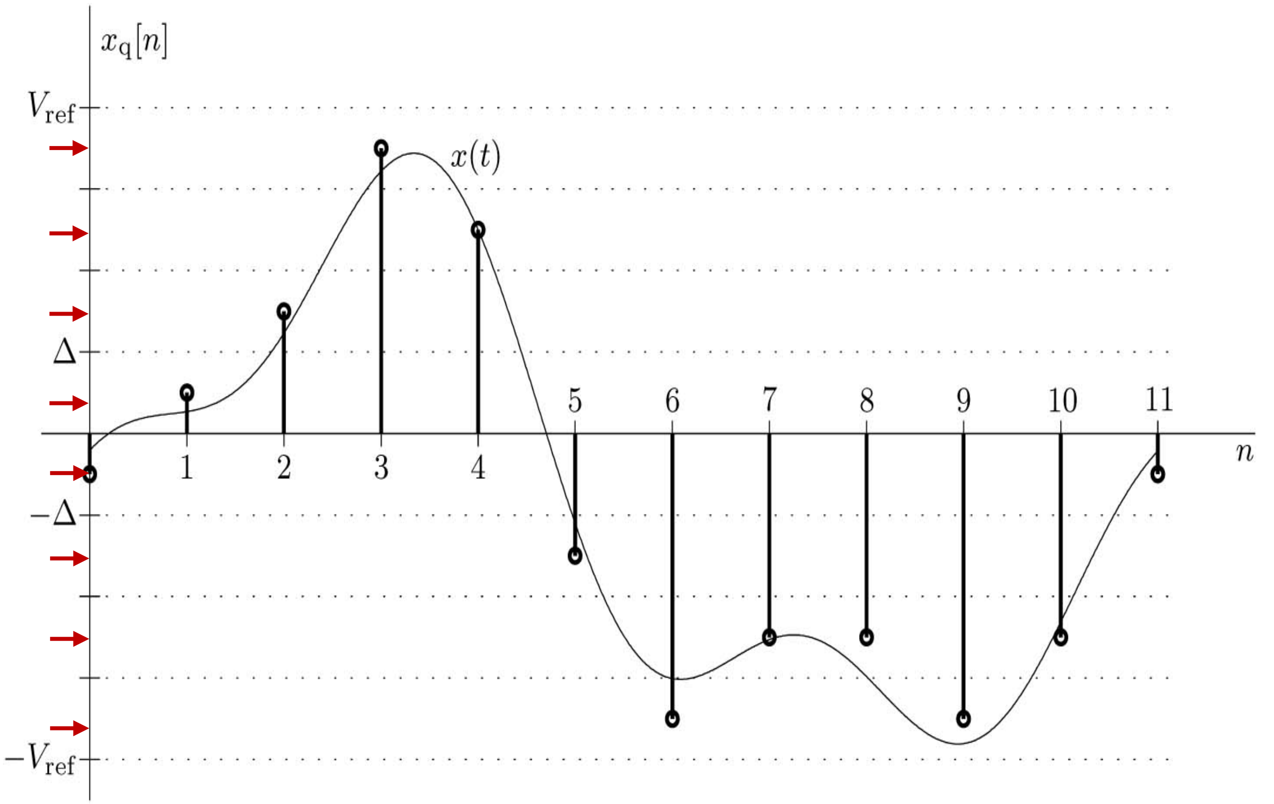
$$L = 2^B \qquad B = \log_2 L$$

$$\Delta = 2\, V_{\text{ref}} / L \qquad \Delta = V_{\text{ref}} / 2^{B-1}$$

$\Delta$ : Quantization level
$L$ : Number of levels between -$V_{\text{ref}}$ and $V_{\text{ref}}$
$B$ : Number of bits for each sample

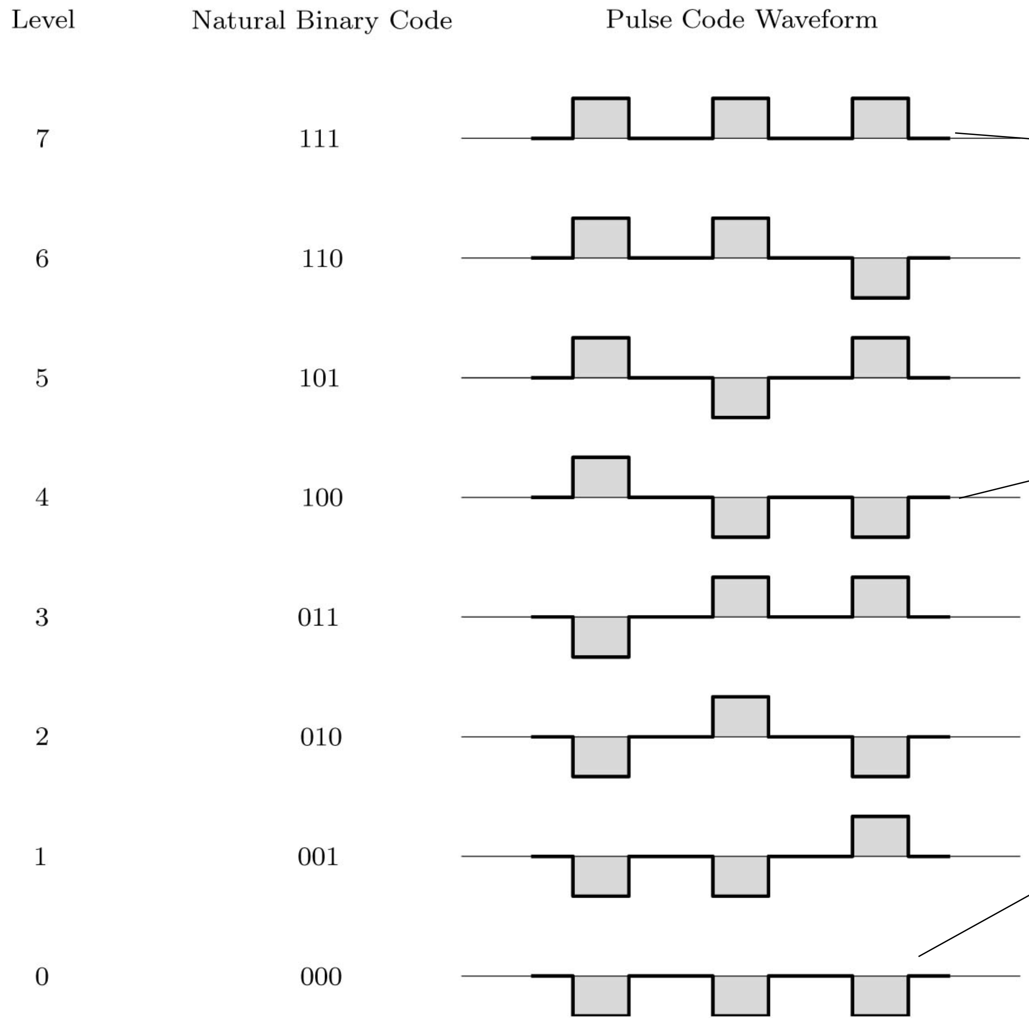
Quantization error $Q_e = \Delta/2$

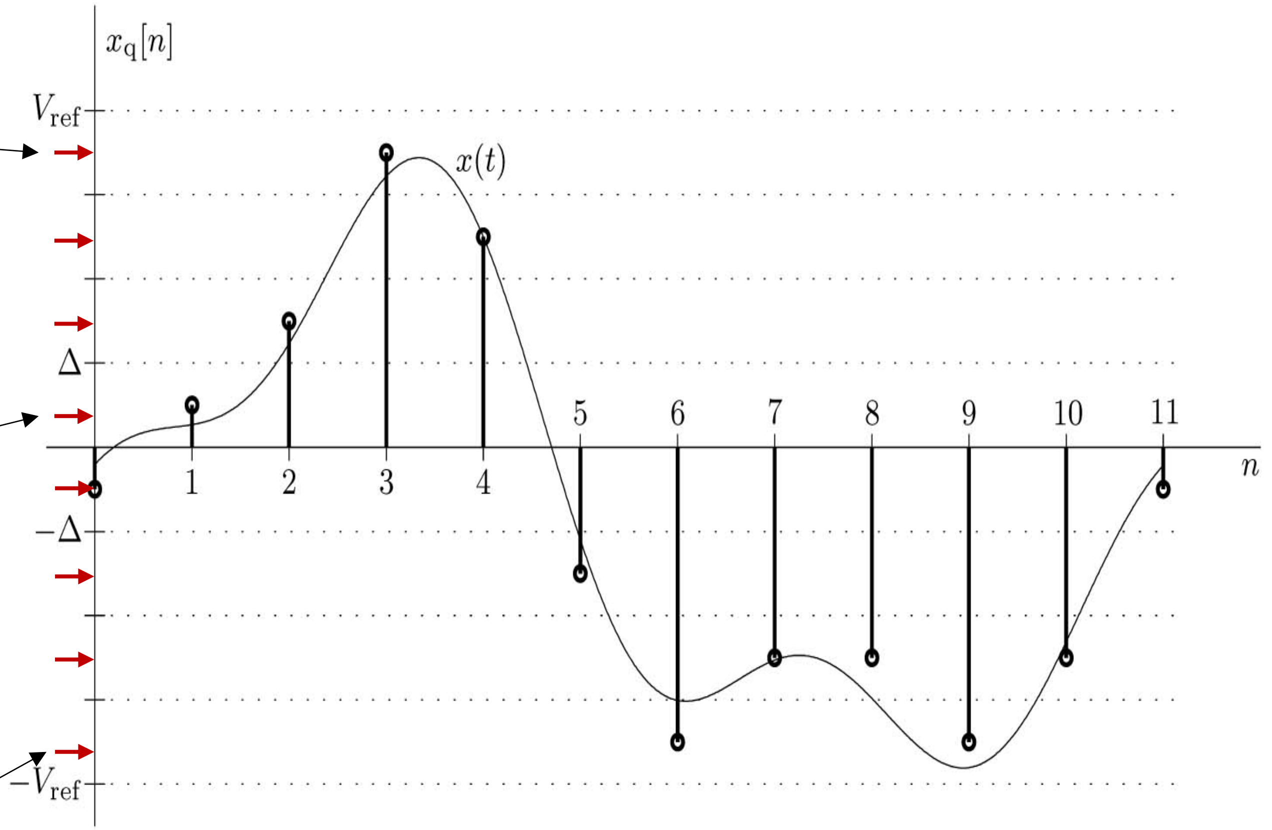
$$Q_e = V_{\text{ref}} / 2^B$$



# 3.6 Analog to Digital Conversion

## Pulse Code Modulation (PCM)




Data Transmission = 3 $F_s$ bits/sec

# Digitizing Voice and Audio

**Telephone Communication:**

Sampling Rate $f_s$ = 8000 samples/sec
Number of bits = 7 bits/sample
Data transmission = 8000*7 = 56000 bits/sec (56 kbps)

**Music:**

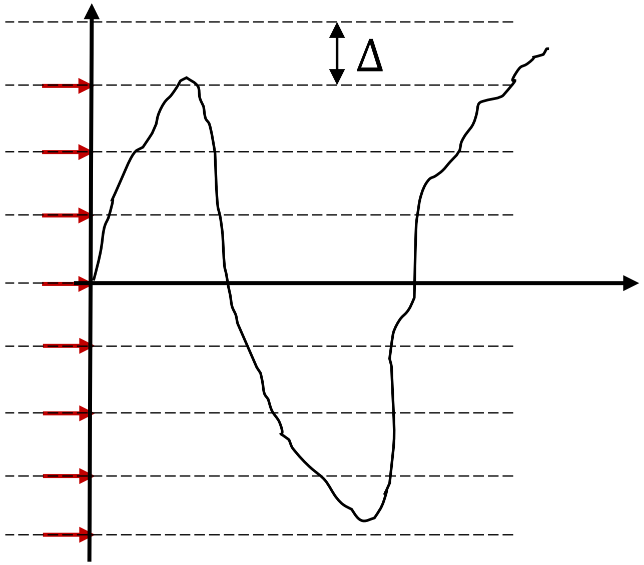
Sampling Rate $f_s$ = 44100 samples/sec
Number of bits = 16 bits/sample
Data transmission = 44100*16 = 1.4 million bits/sec (1.4 Mbps)

**Example**

A signal $x(t)$ bandlimited to 3 kHz is sampled at a rate 30% higher than the Nyquist rate. The maximum acceptable error in the sample amplitude (the maximum error due to quantization) is 0.5% of the peak amplitude $V_{ref}$ . The quantized samples are binary coded. Find the required sampling rate, the number of bits required to encode each sample, and the bit rate of the resulting PCM signal.

# Asymmetric Quantization Converter

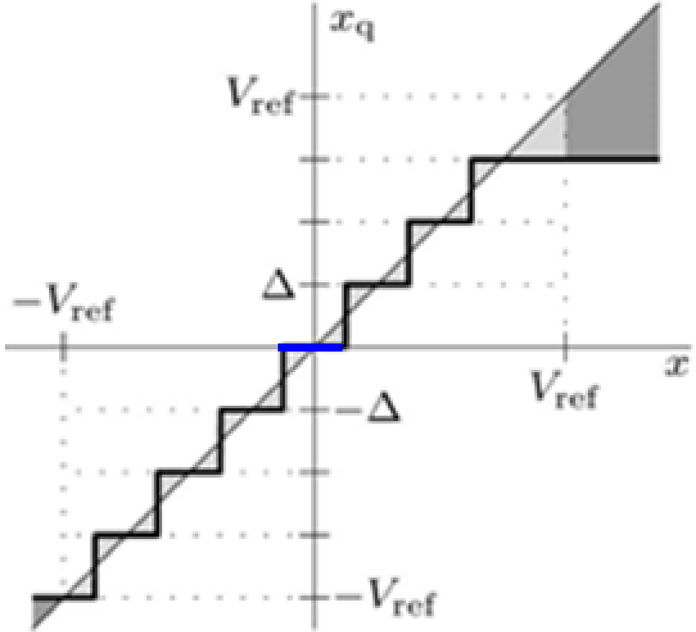
Zero is included and levels above zero does not equals levels below zero.




Rounding Asymmetric

$$x_q = \Delta \left\lfloor \frac{x}{\Delta} + \frac{1}{2} \right\rfloor$$

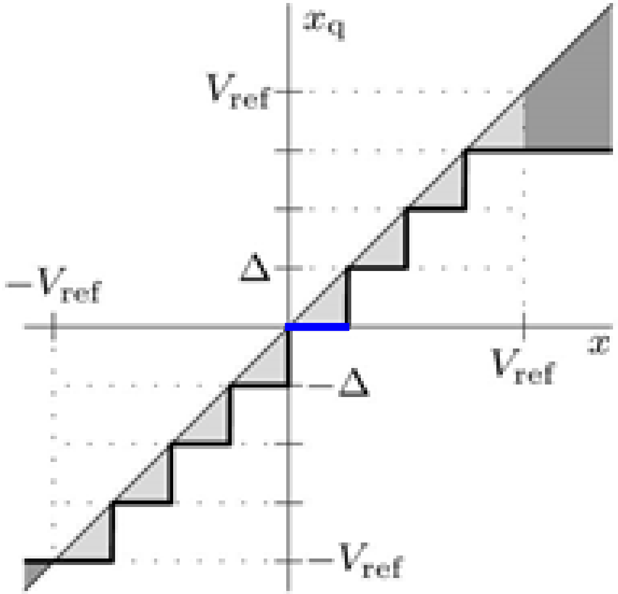
$$x_{\mathrm{q}} = \frac{V_{\mathrm{ref}}}{2^{B-1}} \left\lfloor \frac{x}{V_{\mathrm{ref}}} 2^{B-1} + \frac{1}{2} \right\rfloor$$




Truncating Asymmetric

$$x_q = \Delta \left\lfloor \frac{x}{\Delta} \right\rfloor$$

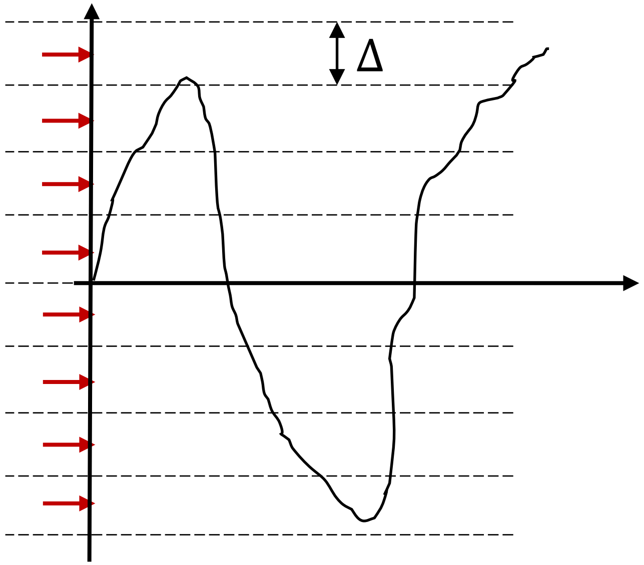
$$x_{\mathrm{q}} = \frac{V_{\mathrm{ref}}}{2^{B-1}} \left\lfloor \frac{x}{V_{\mathrm{ref}}} 2^{B-1} \right\rfloor$$




Find $x_q$ for $x = 8.1$ and $B = 3$bits. $V_{ref} = 10$

# Symmetric Quantization Converter

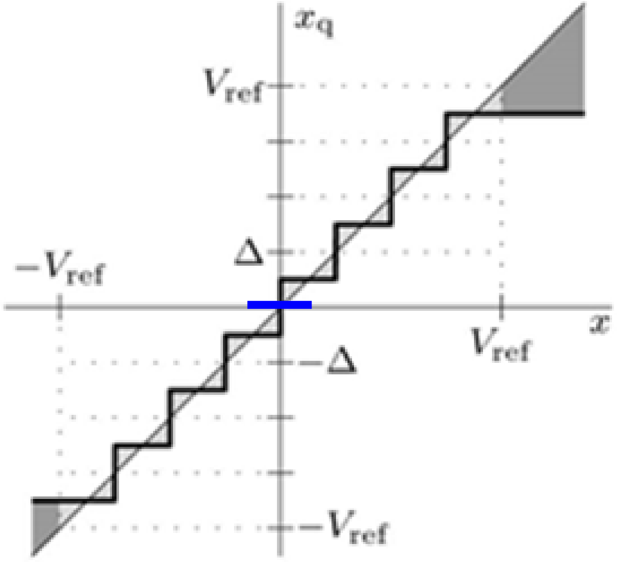
Zero is not included and levels above zero equals levels below zero.




Rounding Symmetric

$$x_q = \Delta\left(\left\lfloor \frac{x}{\Delta} \right\rfloor + \frac{1}{2}\right)$$

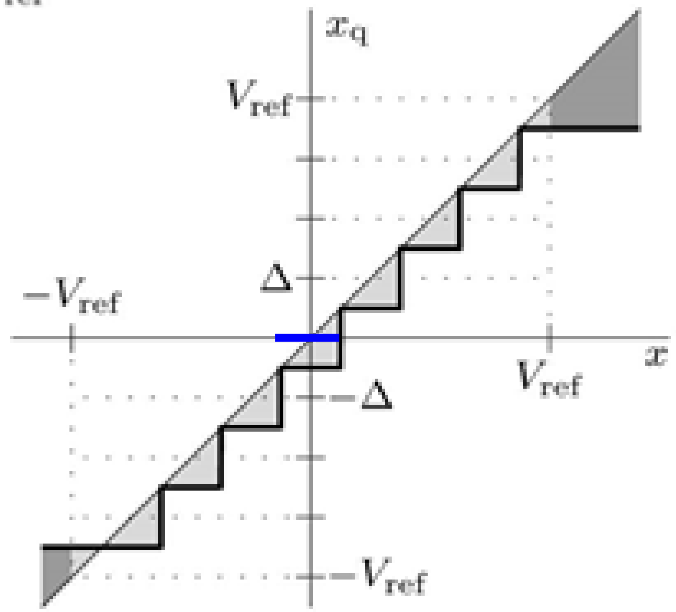
$$x_\mathrm{q} = \frac{V_\mathrm{ref}}{2^{B-1}}\left(\left\lfloor \frac{x}{V_\mathrm{ref}} 2^{B-1} \right\rfloor + \frac{1}{2}\right)$$




Truncating Symmetric

$$x_q = \Delta\left(\left\lfloor \frac{x}{\Delta} - \frac{1}{2} \right\rfloor + \frac{1}{2}\right)$$

$$x_\mathrm{q} = \frac{V_\mathrm{ref}}{2^{B-1}}\left(\left\lfloor \frac{x}{V_\mathrm{ref}} 2^{B-1} - \frac{1}{2} \right\rfloor + \frac{1}{2}\right)$$




Find $x_\mathrm{q}$ for $x$ =7.74 and $B$ = 3bits. $V_\mathrm{ref}$ = 10

## Rounding Asymmetric

$$x_{\mathrm{q}} = \frac{V_{\mathrm{ref}}}{2^{B-1}} \left\lfloor \frac{x}{V_{\mathrm{ref}}} 2^{B-1} + \frac{1}{2} \right\rfloor$$

```
clear all
[sig Fs]=audioread('Song.wav');
Vref = max(abs(sig));
B = 2; Delta = 2*Vref/(2^(B-1));
xqra = Delta*floor(sig/Delta+1/2);
xqra(xqra>Vref-Delta) = Vref-Delta; xqra(xqra<-Vref) = -Vref;
sound(xqra, Fs)
```

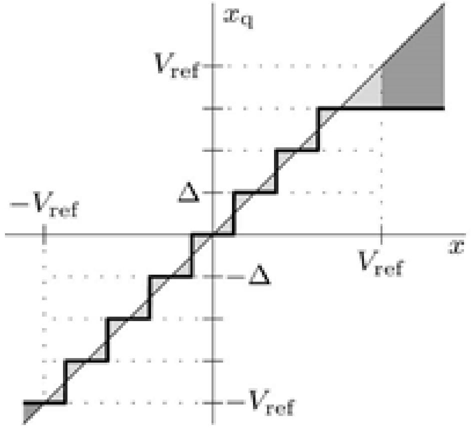
$$x_q = \Delta\left(\left\lfloor \frac{x}{\Delta} \right\rfloor + \frac{1}{2}\right)$$

8 Bits
\# Levels = 256

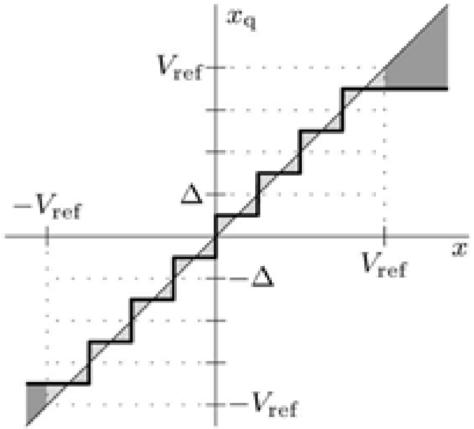
2 Bits
\# Levels = 4

# Example

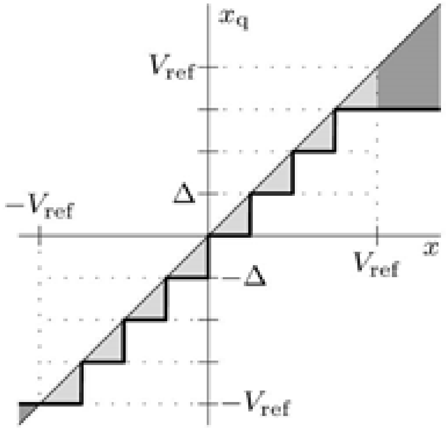
Drill 3.14: Consider the common situation where a signal $x(t)$ becomes 0 for an extended period of time. Explain why an asymmetric rounding converter is better than symmetric rounding converter in the case where low-level noise corrupts $x(t)$. Which asymmetric converter is best, rounding or truncating? Explain.



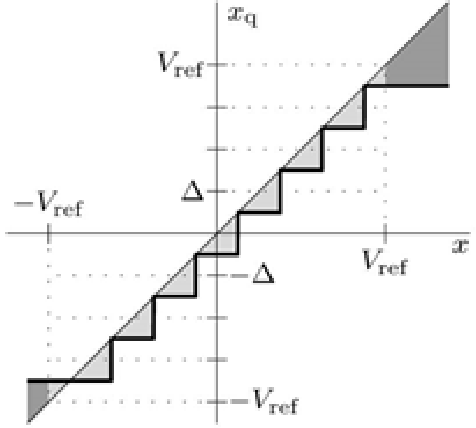

Rounding Asymmetric



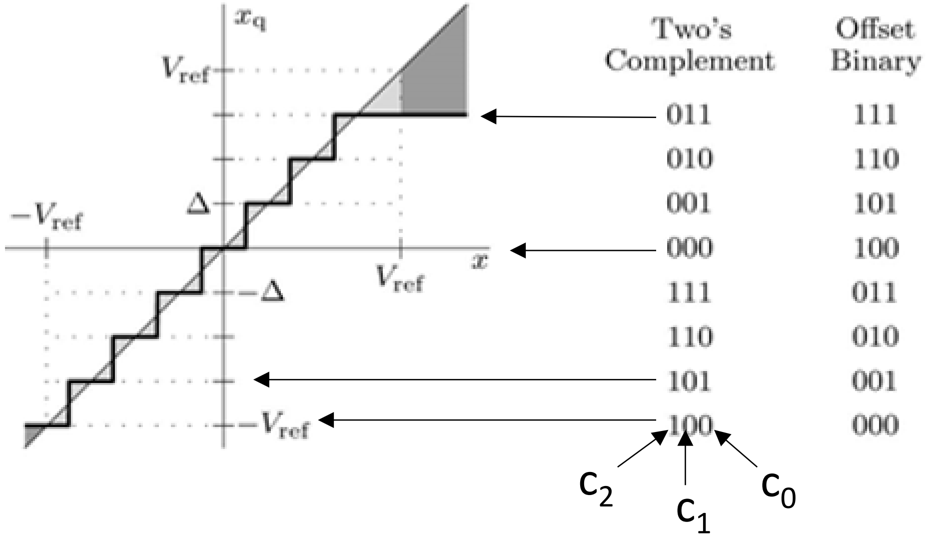

Rounding Symmetric




Truncating Asymmetric




Truncating Symmetric




## Example of Two's Complement

3 → 011

To find the binary for a negative value then reverse the bits of the positive value and add 1.

-3 → 100 + 001 = 101

## decoding Method for asymmetric

Two's Complement $$x_q = V_{\mathrm{ref}}\left(-c_{B-1}2^0 + c_{B-2}2^{-1} + \cdots + c_1 2^{-(B-2)} + c_0 2^{-(B-1)}\right)$$

Offset Binary $$x_q = V_{\mathrm{ref}}\left(-1 + c_{B-1}2^0 + c_{B-2}2^{-1} + \cdots + c_1 2^{-(B-2)} + c_0 2^{-(B-1)}\right)$$

Note: for symmetric add $\Delta/2$ to $x_q$

Example 3.9: Assuming a 4-bit asymmetric converter is used, determine the value $x_q$ that corresponds to the two's complement code word 1101. What is the value if the code word is offset binary? How do the values change if the converter is instead a symmetric converter?

Example 3.10: Consider the signal $x(t) = \cos(2\pi t/25)$. Using sampling interval $T = 1$ s, reference $V_{\text{ref}} = 1$, and $B = 2$ bits, plot the quantized signal $x_q[n]$ over one period of $x(t)$ using rounding asymmetric quantization methods of Eq. (3.23).

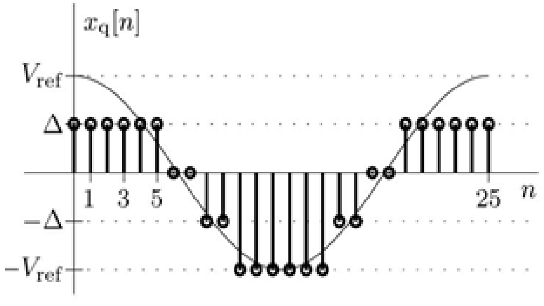
$$x_{\text{q}} = \frac{V_{\text{ref}}}{2^{B-1}} \left\lfloor \frac{x}{V_{\text{ref}}} 2^{B-1} + \frac{1}{2} \right\rfloor$$

## Matlab Code

$$x_q = \Delta \left\lfloor \frac{x}{\Delta} + \frac{1}{2} \right\rfloor$$

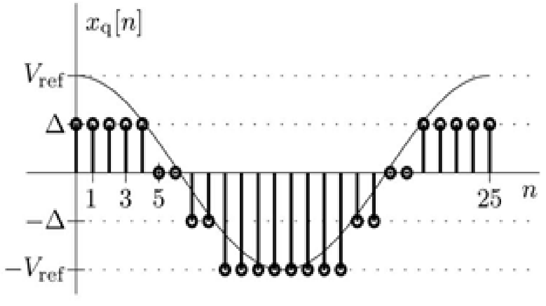
```
T = 1; Vref = 1; B = 2; Delta = 2*Vref/(2^B);
x = @(t) cos(2*pi*t/25); n = (0:25);
xqra = Delta*floor(x(n*T)/Delta +1/2);
xqra(xqra>Vref-Delta) = Vref-Delta; xqra(xqra<-Vref) = -Vref;
stem(n,xqra);
```

Rounding asymmetric



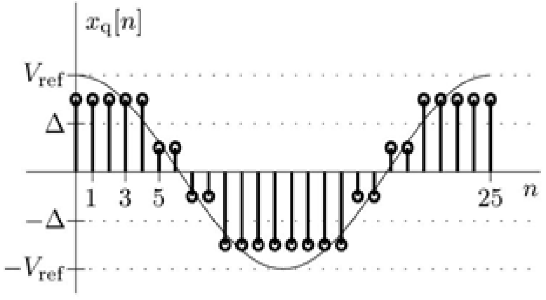

(a)

Truncating asymmetric



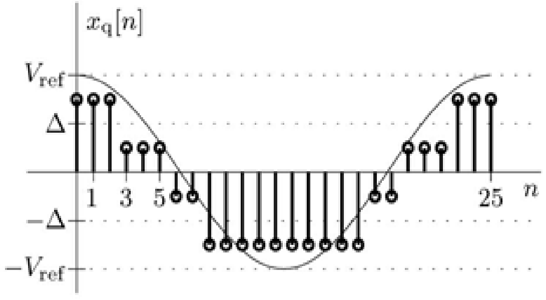

(b)

Rounding symmetric




(c)

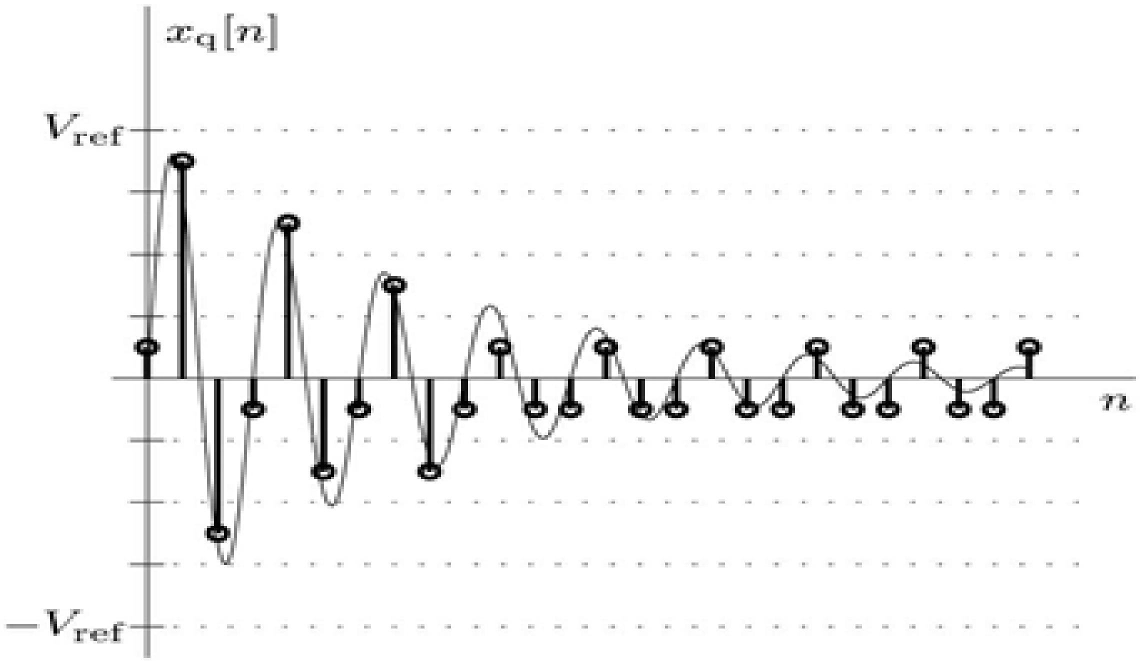
Truncating symmetric



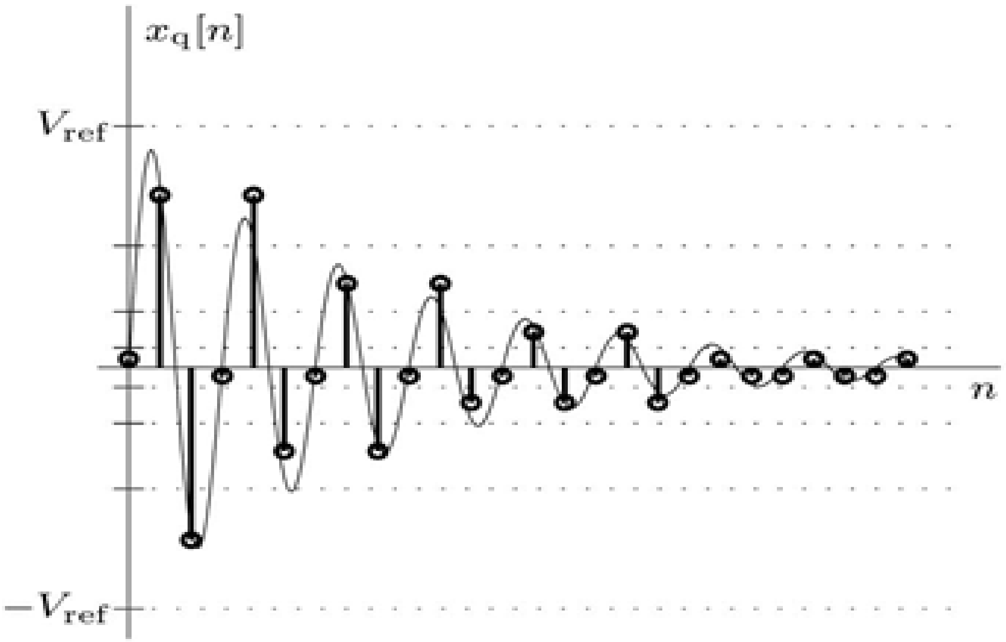

(d)

# Nonlinear Quantization

- Reduce quantization error for smaller signal amplitudes at the expense of larger signal amplitudes.
- In telephone the bit rate for linear quantization is twice that for nonlinear to establish the same sound quality.




Linear quantization




Nonlinear quantization

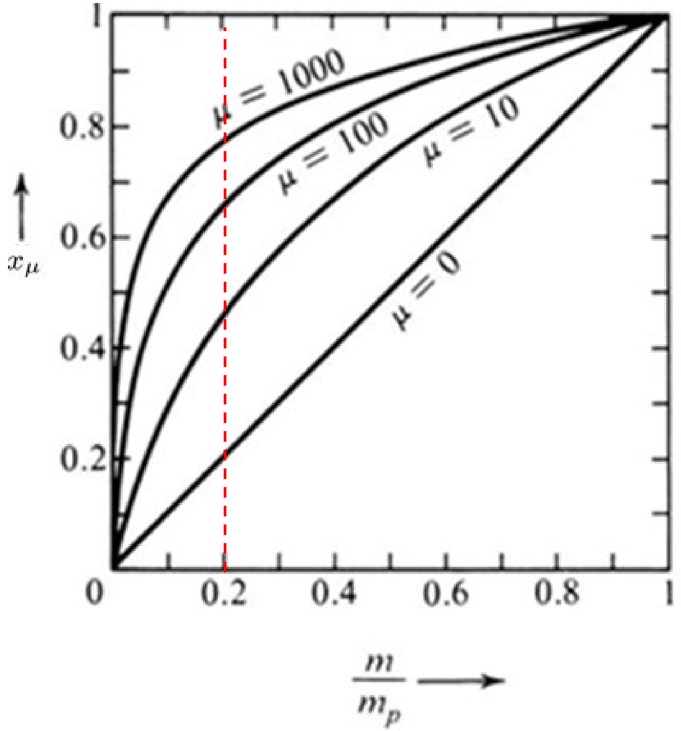
# Nonlinear Quantization

*μ-law Compression* $$x_\mu = \mathrm{sgn}(x)\left[\frac{\ln(1+\mu|x|/V_{\mathrm{ref}})}{\ln(1+\mu)}\right]V_{\mathrm{ref}}$$

The compression parameter $\mu$ determines the level of compression. Large $\mu$ increases resolution for small signal amplitudes, all at the price of more coarseness for large amplitudes.

*Inverse μ-law Compression*

$$x = V_{ref}\,\mathrm{sgn}(x_\mu)\left(\frac{1}{\mu}\right)\left[(1+\mu)^{\lfloor x_\mu/V_{ref}\rfloor} - 1\right]$$

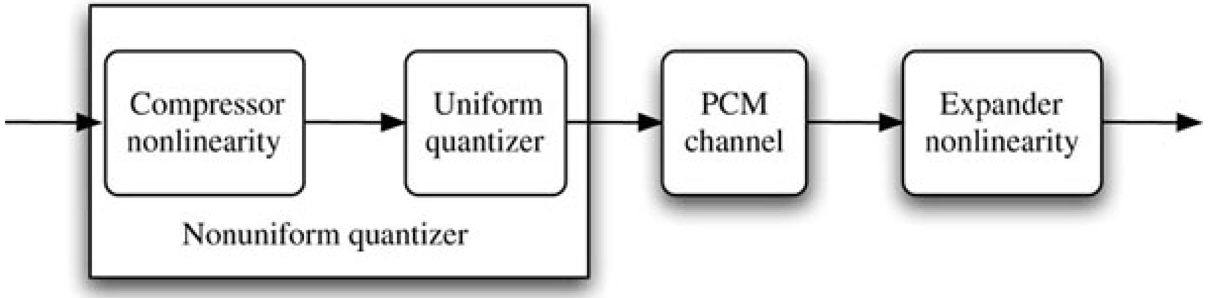



Example

$V_{\mathrm{ref}} = 10 \quad \mu = 100$

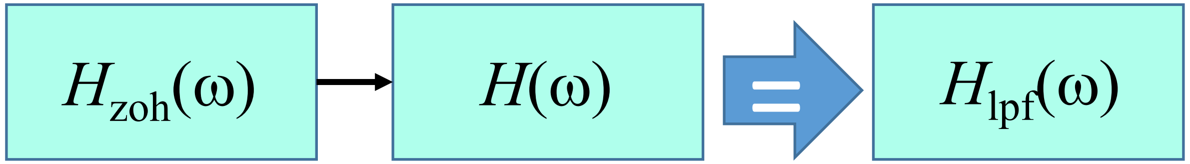
$x_1 = 1 \qquad x_{\mu 1} = 5.196$

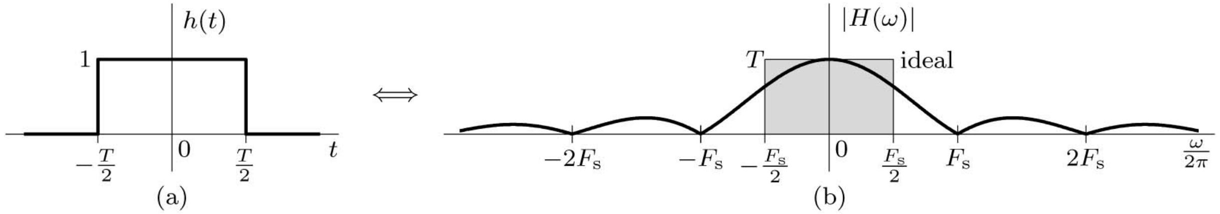
$x_2 = 8 \qquad x_{\mu 2} = 9.52$



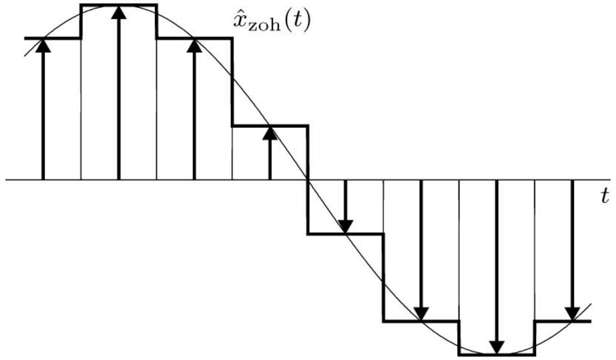
# Source of Distortion in Signal Reconstruction

## 1- Aperture Effect: Distortion from the Zero-Order- Hold



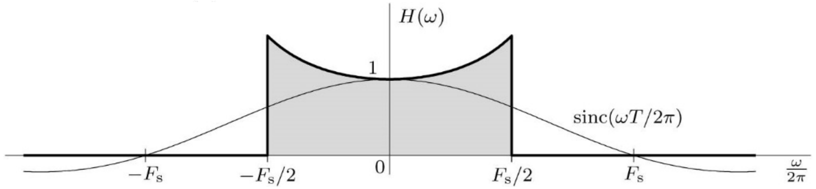




$$\Pi\left(\frac{t}{\tau}\right) \longleftrightarrow \tau \operatorname{sinc}\left(\frac{\tau\omega}{2\pi}\right)$$




$$H_{zoh}(\omega) = T \operatorname{sinc}\left(\frac{\omega T}{2\pi}\right) e^{-j\omega T/2}$$

$$H_{lpf}(\omega) = T\,\Pi\left(\frac{\omega}{2\pi F_s}\right)$$




$$H_{zoh}(\omega)H(\omega) = H_{lpf}(\omega) \qquad H(\omega) = \frac{H_{lpf}(\omega)}{H_{zoh}(\omega)}$$

$$\frac{H_{lpf}(\omega)}{H_{zoh}(\omega)} = \frac{\Pi\left(\frac{\omega T}{2\pi}\right) e^{j\omega T/2}}{\operatorname{sinc}\left(\frac{\omega T}{2\pi}\right)} \qquad \left|H(\omega)\right| = \frac{\Pi\left(\frac{\omega T}{2\pi}\right)}{\operatorname{sinc}\left(\frac{\omega T}{2\pi}\right)} = \begin{cases} \dfrac{\omega T/2}{\sin(\omega T/2)} & |\omega/2\pi| < F_s/2 \\ 0 & |\omega/2\pi| > F_s/2 \end{cases}$$

# Source of Distortion in Signal Reconstruction

## 2- Distortion from Linear Quantization

Quantization error $e_q$ value is in the range (-Δ/2, Δ/2)

Energy (or power) of the quantization error is the means square quantization error $E_q$

$$E_{\mathrm{q}} = \frac{1}{\Delta} \int_{-\Delta/2}^{\Delta/2} e_{\mathrm{q}}^2 \, de_{\mathrm{q}} = \frac{\Delta^2}{12} \qquad \Delta = 2V_{\mathrm{ref}}/L \text{ and } L = 2^B$$

$$E_{\mathrm{q}} = \frac{V_{\mathrm{ref}}^2/3}{4^B}$$

From probability the $e_q$ is a random variable with uniform density function

$$E_{\mathrm{q}} = \int_{-\Delta/2}^{\Delta/2} e_{\mathrm{q}}^2 f(e_{\mathrm{q}}) \, de_{\mathrm{q}} = \int_{-\Delta/2}^{\Delta/2} \frac{e_{\mathrm{q}}^2}{\Delta} \, de_{\mathrm{q}} = \frac{\Delta^2}{12}$$

# Example 3.11

A periodic sinusoidal signal $x(t)$ of amplitude 12 volts is quantized and binary coded. Assuming the power of the signal-to-quantization noise ratio must exceed 20000, determine the smallest number of bits $B$ needed to encode each sample.

Answer: $B \geq 6.8514$ so choose $B = 7$

HW 2 is due